

Ultraintuitionism: the finite as a principal-ultrafilter shadow of the One

Ultraproducts, Łoś transfer, and imprecise construction
for conjecture, credence, and concept formation

Mettapedia Working Notes

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Abstract

Ultraintuitionism holds that only sufficiently small finite objects are real. *Ultraintuitionism* is its inverse: only the infinite is truly real, and the finite is a *simulation*—a coordinate-shadow that a *principal ultrafilter* projects out of the One. (This anti-finitary sense is distinct from, and nearly opposite to, an older use of the word for a programme *beyond* the transfinite; we develop the former, which has lacked a worked mathematical account.) This essay steelmans that thesis and gives it teeth. The phenomenological motivation is direct: nothing finite is ever *given* in experience; every act of looking deeper finds more, granularity never bottoms out, and even the discrete quanta of physics resolve, on inspection, into excitations of continuous fields. The mathematical home is equally direct: the ultraproduct $(\prod_i M_i)/\mathcal{U}$ and Łoś's theorem. A *principal* ultrafilter selects a single coordinate—a finitary shadow; a *free* ultrafilter (the canonical **hyperfilter**) builds a genuinely infinite object—the One; and Łoś's theorem is the law that transfers truth between them. We show that three constructions already *proved*, sorry-free, in the Mettappedia Lean 4 development are exactly the load-bearing instances of this picture: (i) Łoś's theorem itself, native to the repository, gives a semantics for *conjecturing in the limit*; (ii) the credal-width theorems make the *precise = principal*, *imprecise = free* dial mathematically exact; and (iii) the de Finetti machinery realizes the *infinite-primary* construction order in which finite-prefix laws are mere projections of one infinite measure. We separate scrupulously what is proved from what is envisioned, and we draw out the philosophical and practical ramifications—for automated conjecture, for imprecise Bayesianism, and for the discrete/continuous boundary in physics and machine learning.

1 Prologue: nothing finite is ever given

Begin not with axioms but with attention. Try to be shown something finite—a single, complete, bounded thing, given all at once with nothing left over. You will not be shown it. Whatever you fix upon, you can look closer: the edge of the table dissolves into fibre, the fibre into cellulose, the molecule into a cloud of amplitude. At no stage does a final grain appear and announce itself as the last. The phenomenology of perception is a phenomenology of *regress*, not of arrival.

Physics, our most disciplined looking, tells the same story when pressed. The quantum looks like the triumph of the discrete—a smallest packet, a countable click. But the quantum is not a pebble; it is an *excitation of a field*, and the field is a continuous assignment of amplitude over all of space and time. The particle is what a bounded, situated observer *distinguishes* within an unbounded continuum; it is a reading, not a brick. The world-crystal and observer-relative reconstructions of quantum structure make this precise from the physics side: local, conserved, particle-like behaviour *emerges* from a continuum substrate under the regularities an observer can resolve, rather than the continuum being assembled from particles [7, 8].

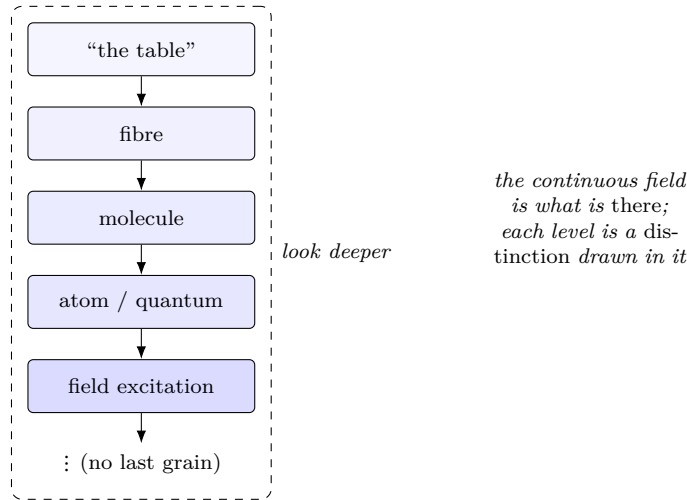


Figure 1: The phenomenological regress (D5). Each finite “object” is a distinction drawn within a continuum that itself never appears as a finite datum. The arrows never terminate in a last grain.

The moral is not scepticism about the finite; finite descriptions are indispensable. The moral is about *priority*. The finite is something we *do*—an act of distinction performed by a situated observer—whereas the infinite continuum is what the doing is done *to*. Ultrafinitism takes this order of priority seriously and asks what mathematics looks like when the infinite is the ground and the finite is the derived, simulated thing.

2 The thesis: the Law of Oneish

Principle 1 (Ultraintinitism). Only the infinite is truly real. Every finite object is a *simulation*: a projection of the One onto a single coordinate, effected by a *principal ultrafilter*. Distinctions—and hence finitudes—are observer-relative carvings of an indivisible infinite whole. Call this last clause the *Law of Oneish*: beneath the carvings there is one thing, and the carvings are how a perspective makes a finite world to live in.

The thesis can be stated as a syllogism. (1) True separation does not obtain in the world—no boundary is absolute, every cut leaks. (2) The world is infinite. *Therefore* (3) no truly finite, fully separated entity exists; finitude is a working abstraction, not a fundamental fact. The number 1, on this reading, has its *formal* home as a *terminal object*: an entity with no internal parts of its own, individuated only by the unique map into it from every object (Section 3.1); the looser gloss “1 as the equivalence class of all objectifiable systems counted as one” is the intuition, not the technical core. In a phenomenological register the same point reads as analytic idealism: finitary distinctions are contortions in an undivided field of awareness, useful folds rather than separate things.

The step from phenomenology to ontology should not be rushed; that “nothing finite is given” supports anti-atomism and observer-relativity more directly than full monism. We therefore separate two bridge principles and grade them honestly.

Principle 2 (No-Atom, from No-Separation). If no boundary is absolute—if every individual leaks into a context it cannot be cleanly separated from—then there is no *atom*: no minimal, fully separated individual. This is the well-supported, anti-atomist core; phenomenology presses it directly. It yields *atomlessness* (Section 6), not yet monism.

Principle 3 (Priority of the Whole). The infinite whole is prior to its finite carvings: a carving presupposes the manifold it carves, but the manifold does not presuppose any particular carving. This is the substantive metaphysical commitment that turns anti-atomism into ultraintinitism proper. We flag it as a commitment, argued for but not forced by the phenomenology alone.

Remark 4 (a note on the term). “Ultraintinitism” has been used in two nearly opposite senses. In a 2012 MathOverflow discussion the word names a *trans-transfinite* programme—a step *beyond* the cumulative hierarchy V , postulating hyper-cardinal or “ultimate-class” size notions larger than any ordinary cardinal, with the hyperuniverse programme, universal-set theories, and $V \rightarrow V$ wholeness axioms as neighbours [19]. The sense developed here is the *anti-finitary* one of the contemporary #UltraInfinitism stance: not bigger infinities beyond V , but the claim that *the finite does not fundamentally exist*. Across the available informal sources this second sense has circulated as a metaphysical slogan rather than a worked mathematical program, against the mature philosophical backdrop of debates over actual versus potential infinity [20]. Supplying such a

program—an ultraproduct semantics with proven anchors and a type-theoretic home—is the contribution of this essay.

This inverts the usual constructional order. Carnap’s *Aufbau* and its modern descendant, Chalmers’s scrutability programme, build the world *up* from a compact base by a priori entailment [1, 2]; Russell’s “supreme maxim” urges us to replace inferred entities with *constructions* out of what is given [15]. Ultrafinitism keeps the constructional spirit but reverses the arrow: the base is not a stock of finite givens from which the infinite is approached, but the infinite One from which the finite is *projected down*. The technical engine of that projection is the ultrafilter, to which we now turn.

3 The mathematical home: ultraproducts, ultrafilters, Łoś

Let $(M_i)_{i \in I}$ be a family of structures and $\mathcal{U}$ an ultrafilter on the index set I . The *ultraproduct* $(\prod_i M_i)/\mathcal{U}$ is the product quotiented by the relation “agree on a $\mathcal{U}$ -large set of coordinates.” Two extremes govern its character.

- A *principal* ultrafilter $\mathcal{U} = \{S : i_0 \in S\}$, generated by a single index i_0 , collapses the ultraproduct to the single factor M_{i_0} . It *simulates the finitary*: the One is read off at one coordinate, and what you get back is exactly one finite shadow.
- A *free* (non-principal) ultrafilter—canonically the **hyperfilter**, extending the cofinite filter on an infinite index set—ignores every finite set of coordinates and builds a genuinely new, infinite object: the *One*. Robinson’s hyperreals are the paradigm: $\mathbb{R}^* = (\prod_n \mathbb{R})/\mathbf{hyperfilter}$, an ordered field with genuine infinitesimals and infinities [14, 9, 10]. In our Lean dependency this identification is literal: the type of hyperreals is defined as `Germ (hyperfilter ℕ) ℝ`.

Remark 5 (the One has genuinely new inhabitants). The free pole is not a re-labelling of the finite. In $\mathbb{R}^*$ the sequence $\omega := [\langle 1, 2, 3, \dots \rangle]$ exceeds every standard real (for each $n_0 \in \mathbb{N}$ the set $\{n : n > n_0\}$ is cofinite, hence in the **hyperfilter**, so $\omega > n_0$), and $\varepsilon := [\langle 1, \frac{1}{2}, \frac{1}{3}, \dots \rangle]$ is a positive infinitesimal with $\omega\varepsilon = [\langle 1, 1, 1, \dots \rangle] = 1$. Yet every first-order truth about $\mathbb{R}$ —the ordered-field axioms, every algebraic identity—transfers to $\mathbb{R}^*$ by Łoś (Theorem 6). This is the Law of Oneish in miniature: the One contains objects no coordinate does, while preserving exactly the first-order shadow each coordinate sees.

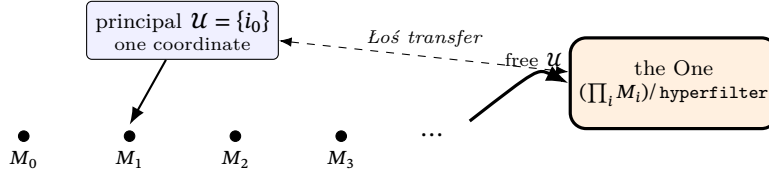


Figure 2: The ultrafilter dial (D1). A principal ultrafilter projects the family to one finite coordinate (a shadow); a free ultrafilter builds the infinite One. Łoś’s theorem is the law that transfers first-order truth across the dial.

The bridge between the two extremes is Łoś’s theorem, which in our repository is proved natively (not imported), sorry-free, as part of a self-contained first-order model theory. Writing $M \models \varphi$ for satisfaction of a sentence:

Theorem 6 (Łoś; `models_Uprod`, `Logic/Foundation/Foundation/FirstOrder/Ultraproduct.lean:121`).

$$\left(\prod_i M_i\right)/\mathcal{U} \models \varphi \quad \Longleftrightarrow \quad \{i : M_i \models \varphi\} \in \mathcal{U}.$$

Read against Principle 1, Łoś’s theorem *is* the Law of Oneish made operational: a property holds of the One exactly when the set of coordinates on which it holds is one the ultrafilter “believes.” For a principal $\mathcal{U} = \{i_0\}$ this reduces to “ φ holds at the single shadow M_{i_0} ”; for a free $\mathcal{U}$ it becomes a statement about the *tail*—about almost all coordinates, in the ultrafilter’s sense of “almost all.” The compactness theorem falls out immediately (`compact`, `Ultraproduct.lean:166`): a theory is satisfiable iff each finite subtheory is, because one assembles a model of the whole from finite models along an ultrafilter. Even consistency, then, is an infinite-primary phenomenon: the infinite model is the witness that the finite fragments cohere.

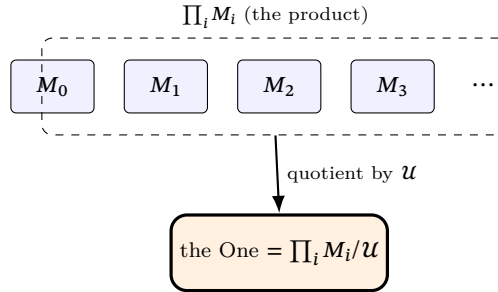


Figure 3: Product-mod-ultrafilter (D2). The finite factors M_i are coordinates; the One is their product collapsed by the ultrafilter’s notion of “almost all.” Nothing finite is fundamental—each M_i is a projection.

3.1 A second home: adjoint modalities and the structural reading

The ultraproduct is one formal home; there is a second, complementary one in type theory, and it answers the natural question “if the finite is a projection, what *kind* of map is the projection?” The answer is: an *adjoint modality*. In Shulman’s cohesive and multimodal adjoint type theories, a continuous (“cohesive”) ground carries a discrete modality $\flat$ that is part of an adjoint string—shape $\mathcal{I} \dashv \flat \dashv \sharp$ codiscrete—so that the *discrete, finitary* layer is *recovered from* the continuous one by a modality with an adjoint, not posited independently [28, 29, 30]. This is the precise type-theoretic form of the intuition that finitary structure is a forgetful/adjoint shadow of a richer continuum, and it is the modal core a multimodal type theory for MeTTa would be built to host.

The same move dissolves the “atom” worry about the number 1. In a structural foundation—an elementary topos, or homotopy type theory—there are no primitive elements at all; 1 is the *terminal object*, characterized purely by a universal property (a unique map from every object), and “elements” of X are maps $1 \rightarrow X$, derived rather than given [12, 27]. The finitary is relational all the way down: “1 as an equivalence class of all objectifiable systems” is exactly “1 as the object every count factors through.”

Remark 7 (the Axiom of Para-Infinity). There is a candid epistemic companion to the metaphysical thesis: we possess *positive evidence for both* “the Universe is finite” and “the Universe is infinite,” so a disciplined reasoner should carry both. This Axiom of Para-Infinity is not a hedge but an instance of Section 5.2: it places a *credal interval* on the finiteness question itself. Where a precise metaphysics would collapse to one principal verdict, Para-Infinity keeps the free ensemble, and the open-world width is the honest report of our uncertainty about the very thing the thesis is about.

4 Ultrafinitism formalized

We can now state the reading precisely, and we index it honestly: there is no canonical free ultrafilter, so there is no single absolute “One.” Given a family $M = (M_i)_{i \in I}$ and a free ultrafilter $\mathcal{U}$ on I , the *relative One* is the ultraproduct

$$\text{One}_{(M, \mathcal{U})} := (\prod_i M_i) / \mathcal{U}.$$

We write $M^* := \text{One}_{(M, \text{hyperfilter})}$ for the canonical-cofinite choice and keep “the One” as evocative shorthand once the indexing is in place. What deserves the definite article is not any one $\text{One}_{(M, \mathcal{U})}$ but the *invariant*: by Łoś (Theorem 6), the *first-order theory* $\text{Th}(\text{One}_{(M, \mathcal{U})})$ depends only on the “ $\mathcal{U}$ -almost-all” filter and is preserved under every coordinate permutation; the dependence on *which* free $\mathcal{U}$ is exactly the perspectival latitude that Section 5.2 measures as credal width. Singular language tracks the invariant; the indexed object tracks the choice.

Definition 8 (envisioned reading). A first-order property φ is *ultrafinitistically true* of the family when it holds in the One, $M^* \models \varphi$. A truth-value assessment is *precise* when it is taken along a principal ultrafilter (a single coordinate) and *open* when it is taken along a free ultrafilter (the One, with its tail-relative verdict).

Three comments fix the status of Definition 8. First, “nothing finite truly exists” is, mathematically, the observation that *no principal ultrafilter is canonical*: there is no privileged coordinate, only the choice of a perspective; the free One is what is invariant under that choice, and Łoś (Theorem 6) is what lets a perspective transfer truth to and from it. Second, Definition 8 is now *built*: the formal core (`Mettapedia/Logic/Metaphysics/UltrafinitismCore.lean`, sorry-free and axiom-clean) defines `UltraTrue` (= *ultrafinitistically true*: truth from the perspective of an ultrafilter, the verdict of the relative One) together with `PreciseFamily/OpenFamily`, and proves the small canonical lemmas this section promises: *principal collapse* (`ultraTrue_pure`: from a principal perspective, ultrafilter truth is truth at the generating coordinate), *invariance* (`ultraTrue_map`: re-coordinatizing the index re-coordinatizes the One and changes nothing else), the *envelope theorems* (`ultraTrue_all_iff/_exists_iff`: the lower/upper envelope over all perspectives is the universal/existential coordinate verdict), the *dial dichotomy* (`openFamily_iff_not_precise`: a family is open exactly when it is not precise, i.e. exactly when the coordinates genuinely disagree), the *free/principal separation* (`hyperfilter_pure_disagree`: the free One affirms what a finite coordinate-shadow denies), and the Łoś *transfer* in wrapper form (`ultraTrue_iff_uprod`, with `ultrapower_elementary`: a structure is elementarily equivalent to every ultrapower of itself). Third—in the interest of honesty—the file currently named `Mettapedia/Hyperseed/Ultrafinitism.lean` is *not* this construction: it is a budget-bounded “perspective” shell (signal classes, effort costs, availability regions) that models a *finitary observer’s access*, the *dual* pole of the present idea, and is a misnomer to be superseded by the ultraproduct reading. We flag this rather than paper over it.

5 Three theorems with teeth

Ultrafinitism would be mere poetry if it floated free of proof. It does not. Three already-proved, sorry-free fragments of the Mettapedia development are precisely the principal/free, finite/infinite picture in action.

5.1 (A) Conjecturing in the limit

A universal mathematical conjecture—“every finite structure of this kind satisfies φ ”—is, on the ultrafinitist reading, a claim about the One. By Łoś (Theorem 6), $M^* \models \varphi$ iff φ holds on an ultrafilter-large set of finite coordinates. This yields a clean trichotomy for an automated conjecturer that enumerates finite witnesses (loops, groups, graphs) by some size parameter:

- *proved*: φ holds at *every* coordinate, hence in every ultraproduct (equational/first-order transfer);
- *refuted*: φ fails *cofinally*, so a counterexample exits the closure and φ fails for ultrafilter-many coordinates;
- *open*: the truth of φ is *ultrafilter-ambiguous in the tail*—it holds on one infinite set of sizes and fails on another.

The inductive leap from “all sampled sizes” to “all sizes” is thus not a hand-wave but the transfer principle: it asks whether the sampled pattern survives to the One.

Our loop-theory conjecturer exhibits the three regimes concretely. Enumerating finite loops by order and mining equational implications, it certifies some candidates as theorems (transfer to every order, hence to the One), and it watches others break *cofinally*: the implication “Moufang $\Rightarrow$ associative” holds through order eleven and then fails at order twelve—the smallest non-associative Moufang loop—so it fails for ultrafilter-many orders and is refuted in the One. The genuinely interesting residue is the third regime: a hardened candidate such as “flexible and left-alternative $\Rightarrow$ left-inverse-property” survives every order the prover can reach, leaving its status in the One open—an ultrafilter-ambiguous tail awaiting either a transfer proof or a cofinal counterexample. These are not three labels we impose; they are the three ways a sampled pattern can relate to the limit object.

Our concept-formation layer already exhibits the elementary form of this gap. In the formal-concept setting, the closure of a premise over a *sampled* population can strictly exceed its closure over the *full* population, and the difference is a witnessable object. The control example is deliberately homely: over the two-element sample {robin, bat}, the implication “winged $\Rightarrow$ flies” *holds*; over the full context (which contains the penguin) it *fails*.

Theorem 9 (sampled-vs-full frontier; `flies_mem_winged_scrutability_frontier`, `LoopConjectureScrutability.lean:397`).

$flies \in \text{sampledClosure}(\{robin, bat\}, \{winged\})$ and $flies \notin \text{fullClosure}(\{winged\})$.

This is exactly an ultrainfinitist tail phenomenon in miniature: a truth of the finite shadow that the larger (eventually infinite) context overturns. The *credal* version makes the imprecision quantitative, and it is the subject of (B).

5.2 (B) Precise = principal, imprecise = free

Index a family of admissible perspectives by “gates” Γ . Each gate yields a sharp verdict; the ensemble yields an interval. Two theorems make the dial exact.

First, imprecision *is* disagreement among perspectives. The credal width collapses to zero exactly when all gates agree—i.e. exactly when the assessment is, in effect, principal (a single coordinate’s verdict suffices):

Theorem 10 (precision = unanimity; `inheritanceCredalWidth_eq_zero_iff`, `ExtensionalIntensionalDivergence.lean:488`).

$$\text{credalWidth}_\Gamma(\text{sub}, \text{super}) = 0 \iff \forall g, h \in \Gamma : \text{strength}_g(\text{sub}, \text{super}) = \text{strength}_h(\text{sub}, \text{super}).$$

Second, the interval endpoints are not ad hoc: the lower and upper inheritance strengths are the *greatest lower bound* and *least upper bound* of the ensemble of per-gate strengths—the Walley lower and upper envelopes of the induced credal set [16, 17]:

Theorem 11 (Walley envelope; `lowerInheritanceStrength_isGLB` / `upperInheritanceStrength_isLUB`, `InheritanceIntegration.lean:125,139`). `lowerΓ` is IsGLB and `upperΓ` is IsLUB of $\{\text{strength}_g : g \in \Gamma\}$.

The ultrainfinitist reading is then forced. A *principal* perspective collapses the credal set to a point—a precise probability, the special case. A *free* perspective keeps the whole ensemble and reports the envelope—an interval whose width measures how far the perspectives spread, i.e. the tail-ambiguity of the One. Precision is the principal-ultrafilter shadow of an irreducibly imprecise, free-ultrafilter ground. Concept formation shows the same structure with a binary outcome: the credal scrutability gap (concepts formed under some gate but not all) carries envelope width exactly 1 on the gap and 0 off it.

Theorem 12 (gap = unit width; `globalEnvelopeWidth_..._eq_indicator_credalScrutabilityGap`, `LoopConjectureScrutability.lean:330`). `credalScrutabilityGap` = `upperFamily \ lowerFamily`, and the global envelope width of the concept-formation gamble is the indicator of this gap.

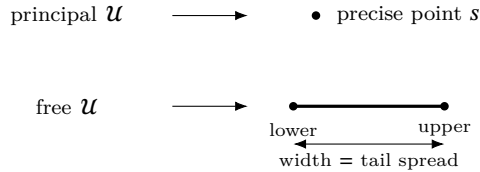


Figure 4: The precise/credal dial (D3). A principal ultrafilter collapses the credal set to a precise point (Theorem 10); a free ultrafilter reports the Walley envelope (Theorem 11), whose width measures perspective spread.

5.3 (C) Infinite-primary de Finetti

The third fragment is the probabilistic heart, and it shows ultrainfinitism not as a reinterpretation but as a *method*. De Finetti’s theorem concerns infinite exchangeable sequences: an exchangeable law on $\{0,1\}^{\mathbb{N}}$ is a mixture of i.i.d. Bernoulli laws, $\nu = \int_{[0,1]} \text{Bernoulli}(\theta)^{\otimes \mathbb{N}} d\mu(\theta)$ [4, 3]. In our development the mixture is a first-class object,

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structure BernoulliMixture with a mixingMeasure on [0,1]
  (Logic/DeFinetti.lean:130),
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and the exchangeable, count-only character of its finite marginals is proved (`infiniteExchangeable_same_counts_same_prob`, `DeFinetti.lean:203`); the unique latent measure is exported stably (`CategoryTheory/DeFinettiStableExports.lean:31`).

The ultrafinitist point is about *construction order*. The naive impulse is to build the global object by *extending* finite-prefix data outward to the infinite sequence—and this is exactly where such constructions stall, on the gap between finitely-specified prefix laws and a coherent law on the full product. Ultrafinitism says: do not extend the finite; *the infinite is primary*. Construct ν on $\{0,1\}^{\mathbb{N}}$ first—as a genuine infinite-product measure, the probabilistic sibling of the ultraproduct—and recover each finite-prefix law as a *projection* (a marginal), not as a building block. The “raw versus finite” gap dissolves because the arrow was pointing the wrong way: the One measure is the datum; the finite prefixes are its principal-ultrafilter shadows.

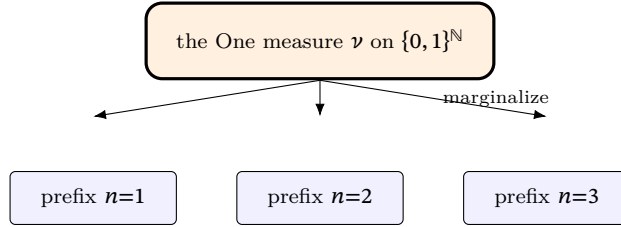


Figure 5: Infinite-primary de Finetti (D4). The infinite mixture ν is the datum; finite-prefix laws are downward projections (marginals), not building blocks.

6 Native homes, and a worked prototype: gunky mereology

The ultraproduct core says *what* the finite is (a principal shadow) but not *where* an infinite-native universe should live—a foundation in which “no finitary elements” is built in rather than asserted against a finitary scaffolding (Section 9). Four candidate homes recur; we score them and then prototype the most tractable.

Native home	no empty individual	no atomic base	observer- relativity	constructive friendly	Lean feasibility
Universal-set (NF/NFU)	no	partial	weak	no	<i>Con</i> formalized ¹
Non-well-founded (AFA)	no	yes	partial	partial	core in mathlib
Atomless gunk	yes	yes	yes	yes	prototyped here
Structural (topos/HoTT)	yes	yes	yes	yes	topos in mathlib; HoTT elsewhere

Table 1: The native-foundation scorecard. Gunk most directly answers the empty-set/atom issue while staying closest to the phenomenology, and it is the one we prototype below.

6.1 Gunk forces the infinite

A *gunky* (atomless) order is one in which every individual is properly divisible—the mereological form of “you can always look deeper.”

Definition 13 (gunky order). A bounded order $(\alpha, \leq, \perp)$ is *gunky* when every non-bottom element has a proper non-bottom part: $\forall a \neq \perp, \exists b \neq \perp, b < a$. Equivalently, α has *no atoms*.

The load-bearing fact—and the formal counterpart of Bridge Principle 2 feeding into Principle 3—is that gunk *forces* infinitude. The finite cannot be gunky.

Theorem 14 ($\text{gunk} \Rightarrow \text{infinite}$; `infinite_of_isGunky`, `GunkyMereology.lean`). *Every gunky order with more than one element is infinite. Iterating “take a proper non-bottom part” produces a strictly descending sequence $a_0 > a_1 > a_2 > \dots$, an injection $\mathbb{N} \hookrightarrow \alpha$.*

Gunk is not vacuous, and it is not universal—we give both poles. The two genuinely two-dimensional algebra witnesses are the ones one wants:

Theorem 15 (two gunky algebras; `isGunky_clopens_cantor`, `isGunky_opens_real`). *The Boolean algebra of clopen subsets of Cantor space $2^{\mathbb{N}}$ is gunky (a nonempty clopen splits: fix one more coordinate), and the frame—complete Heyting algebra—of open regions of $\mathbb{R}$ is gunky (shrink the radius). The non-negative reals furnish the linear case (every magnitude is halvable), while a two-valued algebra is the paradigm non-gunky one (`not_isGunky_bool`): its top is an atom.*

All of Theorems 14–15 are formalized in Lean 4, sorry-free and with only the standard axioms; the statements are transcribed in Appendix A. Spatial regions of the continuum, and clopen cylinders of Cantor space, are thus genuine atomless universes—no empty individual is forced, no minimal part exists, and the universe is provably infinite.

¹The consistency of New Foundations is verified in Lean 4 (Holmes–Wilshaw), so the universal-set home is now machine-checked at the consistency level [32]; it remains a heavy-weight, separate development.

6.2 Stone duality: atoms are the principal-ultrafilter points

Gunk connects to the paper’s central dial not by analogy but by Stone duality, which is the place the principal/free story comes closest to a representation theorem (answering, in part, the critique that the dial is only an organizing principle).

Theorem 16 (Stone duality: atomless $\Leftrightarrow$ perfect). *Let B be a Boolean algebra and $S(B)$ its Stone space, realized as the prime spectrum of the Boolean ring of B . Then there is an order isomorphism $B \cong \text{Clopens}(S(B))$ (the Stone representation: B is the clopen algebra of its Stone space), $S(B)$ is compact, Hausdorff and totally separated, and the atoms of B correspond exactly to the isolated points of $S(B)$ —which, in the Stone topology, are exactly the principal ultrafilters. Hence*

$$B \text{ is atomless (gunky)} \iff S(B) \text{ is perfect (has no isolated point)}.$$

In the powerset $B = \mathcal{P}(X)$ the atoms are the singletons $\{x\}$, i.e. the supports of the principal ultrafilters `pure x` (`isAtom_set_singleton`); so $\mathcal{P}(X)$ is never gunky for inhabited X (`not_isGunky_set`).

This is now formalized *in full generality*—for an arbitrary Boolean algebra, not merely the powerset or Cantor instances—sorry-free and axiom-clean: `stoneRepr` (the order isomorphism) and `isGunky_iff_perfect_stoneSpace` (the equivalence) in `Mettapedia/Logic/StoneGunkDuality.lean`; the bottleneck, that the Boolean-ring spectrum is Hausdorff, is proved from the complementary clopens `basicOpen e` and `basicOpen (1 - e)`. Read against the dial: an atom is precisely a *principal-ultrafilter* point, so “gunky” is exactly “no finitary, principal-shadow point stands alone.” The finite (isolated/principal) is never fundamental; an atomless universe has only free, limit points—the relative Ones of Section 4. The slogan “finite = principal shadow” is thereby upgraded to the dual *theorem* “finite atom = isolated point.”

6.3 A model of ultrainfinitism

We can now say what a *model* of the thesis looks like—a structure in which it holds, in the sense in which one builds a model of a set theory.

Definition 17 (gunky–Stone model). A *gunky–Stone model* is a pair $\langle B, S(B) \rangle$ with B an atomless Boolean (or Heyting) algebra and $S(B)$ its Stone space. Reality is B : every individual is infinitely divisible (gunk), and the universe is infinite (Theorem 14). A “finite world a perspective lives in” is a point of $S(B)$ —a choice of ultrafilter; but since B is atomless, $S(B)$ is perfect (Theorem 16), so *no such point is isolated*: every finitary reading is a limit of others. That is the Law of Oneish, modelled.

The Cantor witness is the self-dual jewel: $S(\text{Clopens}(2^{\mathbb{N}})) = 2^{\mathbb{N}}$ itself, which is perfect. The gunky algebra and its perfect point-space are two

faces of one object, and “no isolated point” is “no finite atom.” We have built the universe (two atomless witnesses), the no-atom axiom (gunk), the infinitude theorem, the Stone representation $B \cong \text{Clopens}(S(B))$, and—now in full generality—the classical equivalence $\text{atomless} \Leftrightarrow \text{perfect}$ (Theorem 16). The point-layer’s Łoś transfer is now also in place: the formal core (Section 4) provides `ultraTrue_iff_uprod`—ultrafinitist truth of a first-order sentence along a family *is* truth in the ultraproduct—and its constant-family corollary `ultrapower_elementary`: a structure is elementarily equivalent to every ultrapower of itself, so B and the Ones built over it validate exactly the same first-order truths. One caveat is layer-honesty, learned from the concept-formation side: the atomic/gunky dial must always be indexed to *which* algebra carries it—a derived algebra (a frontier, a quotient, a concept state) may sit at a different pole than its base. The transfer lemmas (`isGunky_congr`, `isGunky_Iic`, `perfectSpace_stoneSpace_congr`) exist precisely to move the dial to the right carrier.

6.4 Formal philosophy: gunk versus mereological nihilism

The atomless/atomic contrast has a sharp philosophical edge. *Mereological nihilism* holds that there are no composite objects: every object is a mereological *simple* (an atom). Sider [24] argues against it from the mere *possibility* of gunk: (1) nihilism is non-contingent (necessarily true or necessarily false); (2) gunk is metaphysically possible; (3) if gunk is possible, nihilism is not necessary; (4) therefore nihilism is necessarily false.

We formalize the argument with the standard semantic reading of metaphysical modality: a *world* is a parthood structure (a bounded order on ≥ 2 individuals), with $\Box P := \forall W, PW$ and $\Diamond P := \exists W, PW$ (the universal, *S5* frame). *Sider’s nihilism* at a world says “every object is a simple,” $\forall a \neq \perp, \text{IsAtom } a$ (`SiderNihilism`); “gunky” reuses `IsGunky`. Two halves are then *theorems*, sorry-free and axiom-clean, in `Mettapedia/Logic/Metaphysics/SiderNihilism.lean`:

- **The argument is valid, and premises 2–3 are proved.** Gunk is possible (witnessed by the continuum, `possible_gunk`), and a gunky world is never a nihilist world (`gunky_not_siderNihilism`: gunk means *no* atoms, nihilism requires *all* simples). Hence `siders_argument`: $\text{NonContingent}(N) \Rightarrow \Box \neg N$ —granting premise 1, nihilism is necessarily false.
- **But premise 1 fails in the unrestricted logical space of mereologies** (`not_nonContingent_siderNihilism`): there is a gunky world *and* a nihilist world (a two-element order), so neither nihilism nor its negation is necessary. Nihilism is genuinely *contingent* across parthood structures (`siderNihilism_contingent`).

The formalization thus vindicates Sider’s premises 2–3 as theorems and localizes the whole dispute on premise 1: the argument is *valid* but sound only

under the metaphysical posit that restricts the space of worlds so nihilism becomes non-contingent. Via Theorem 16 this is a Stone-dual picture: nihilism is the *discrete* pole (every point of $S(B)$ isolated), gunk the *perfect* pole (no isolated point), and both poles are realized—which is precisely why premise 1 is not a logical truth. The ultrainfinitist takes the gunky/perfect pole as the real one, and `infinite_of_isGunky` with Theorem 16 say why it is the anti-nihilist limit: no isolated point, no finite simple, the universe infinite.

7 Philosophical ramifications

Process before substance. If the finite is an act of distinction rather than a stock of given things, then *becoming* is prior to *being*—a thesis with a long process-philosophical lineage (Whitehead) and a contemporary operational echo in the Hyperseed ontology, where “process” and “becoming” are primitive and a transition phase is explicitly a zone in which the A/B distinction blurs [18, 6]. The ultraproduct gives this teeth: the finite coordinate is the snapshot, the One is the ongoing whole, and Łoś says which snapshot-truths are truths of the whole.

Distinction is observer-relative. The choice of principal ultrafilter *is* the choice of which coordinate to call “the world.” This is the formal residue of the phenomenological point of Section 1: a finite object is a reading, and a reading presupposes a reader—a perspective that draws the distinction. Goertzel’s constructive-nonstandard concept makes the dual movement explicit from the finite side: a finite distinguishing system, with a distinguishability relation DS and a resolution band $[\min_S, \max_S]$, calls *potentially infinite* whatever exceeds its resolution [6]. That is the coordinate’s view looking *up* at the One; ultrainfinitism is the One’s view looking *down* and projecting finitudes. The two are the principal and free poles of a single ultrafilter lattice, and Łoś is the bridge between them.

The continuum as ground. Reading discreteness as emergence rather than foundation aligns ultrainfinitism with the field-theoretic picture in physics and with the internal-language view of topos theory, in which the locally-discrete and the globally-continuous are two faces of one sheaf-theoretic structure [12]. The neutral-monist intuition—that the apparent plurality of things is the differentiation of one underlying stuff [15]—is, on this reading, the Law of Oneish stated in the key of metaphysics rather than model theory.

A caution, in the same spirit of honesty as Section 4: of the thinkers invoked here, only the operational, locally-grounded fragments enter our formal claims. Whitehead, Russell, and Carnap are cited as genuine intellectual lineage, not as authorities whose theses we have formalized.

8 Practical ramifications and applications

Conjecture confidence as tail behaviour. Section 5.1 turns an automated conjecturer’s verdicts into statements about the One. A practical ranking follows: an open conjecture’s credal width (Theorems 10–12) is a measure of *tail ambiguity*, so conjectures whose sampled support is robust across perspectives (small width) are better bets than those that hold only on a perspective-dependent set of sizes. This is a principled, proof-anchored alternative to ranking by raw sample frequency.

Build the infinite first (imprecise Bayes). Section 5.3 is a constructive recommendation, not only a reinterpretation: when a global stochastic process is wanted from a consistent family of finite-dimensional laws, construct the infinite-product/mixture object and read off finite marginals, rather than struggling to extend finite data outward. The imprecise version is the *imprecise Dirichlet* stance [17]: carry a credal set of mixing measures, and let the Walley envelope (Theorem 11) report the free-ultrafilter spread; precision is recovered only by the principal collapse to a single mixing measure.

Scale-relativity in physics and learning (a conjectural prompt). We mark this lane explicitly as suggestive alignment rather than a proved anchor. The renormalization-group picture in physics and the continuum limits of large neural networks *appear* to share the ultrainfinitist shape: the “physical” or “trained” object reads as a scale-relative projection of an underlying continuum, with the finite, discretized model as the coordinate one happens to compute in. That suggests reading finite-size artefacts (lattice spacing, network width) as perspective parameters rather than approximations to apologize for. We offer this as a research prompt for a future toy model, not as evidence on the level of Sections 5–6; the formal core should be hardened first.

9 Objections and replies

A thesis worth holding is worth attacking. We state the strongest objections we know and answer them.

“This is just nonstandard analysis dressed in metaphysics.” Nonstandard analysis is a tool; ultrainfinitism is a claim about *priority* and a *unification*. The tool says infinite and infinitesimal elements can be constructed and reasoned with via ultrapowers; the thesis says the infinite construction is the primary thing and the finite is its projection, and—this is the content beyond NSA—that *the same dial* governs conjecture (Section 5.1), credence (Section 5.2), and concept formation (Theorem 12). What NSA leaves as a technique for analysis, ultrainfinitism reads as a single construction principle across three activities we usually keep apart.

“Free ultrafilters are not canonical—so ‘the One’ is ill-defined.” There are many free ultrafilters and (under the usual axioms) no way to name one; the ultraproduct depends on the choice. We embrace this rather than resist it. The *absence* of a privileged coordinate is exactly Principle 1: no finite shadow is the world. And the residual dependence on *which* free ultrafilter is not a defect but the formal seat of imprecision: it is the tail-ambiguity that Theorem 10 measures as nonzero credal width. What is invariant—the first-order theory transferred by Łoś, the envelope of verdicts—is what ultrainfinitism calls real; what varies with the ultrafilter is exactly what it calls perspectival. The objection, pressed, becomes the theory.

“Skolem and Löwenheim—Skolem show the infinite is not even pinned down.” Correct, and unneeded. Ultraintuitionism does not require a unique infinite model, a categorical theory, or a settled cardinality. It requires only that the infinite be *primary* and the finite a projection, with transfer as the invariant bridge. Non-categoricity is, again, perspectival latitude—more One than any coordinate can pin, which is the expected shape of a ground that exceeds its shadows.

“Then drop first-order logic: full second-order semantics restores categoricity, and your latitude evaporates.” The strongest form of the previous objection, and the most instructive. With *standard* (full) second-order semantics, Dedekind pins $\mathbb{N}$, Huntington pins $\mathbb{R}$, and Zermelo quasi-pins the set universe; Löwenheim–Skolem is a first-order artifact. But by Lindström’s theorem [23], the price is mandatory: any logic with compactness and downward Löwenheim–Skolem is no stronger than first-order, so full second-order logic forfeits completeness, compactness, and any effective proof calculus—it pins the model from a vantage no finite agent occupies (it “decides” the continuum hypothesis without telling anyone how). Henkin’s general models [21] restore the finitary virtues for the *same* syntax, and thereby re-import the latitude; Farmer counts this dual semantics among simple type theory’s virtues [22]. Our reply: the standard/Henkin switch *is* the principal/free dial, reappearing at the level of semantics itself. Full semantics is the ground-side face (categorical, no effective access); the Henkin family of general models is the shadow-side face (effective, many perspectives); and full-semantics categoricity is *borrowed determinacy*— $\mathbb{N}$ is pinned only relative to a metatheoretic totality of “all subsets” that is itself only quasi-categorical. The latitude is not eliminated but promoted one level, which is the thesis exhibiting itself. This reply is now formalized: one monadic second-order syntax with both semantics, in which the ultraintuitionist core theory—gunk, nontriviality, and the genuinely second-order axiom *every ultrafilter is free*—is satisfiable under the standard reading and over *every* Henkin family whatsoever *by the same gunky witness*, the Cantor clopen algebra (`T_UI_satisfiable_standard`, `T_UI_satisfiable_all_henkin`); the free-ultrafilter axiom is provably *equivalent* to first-order atomlessness over standard semantics (`sat_freeUFax_iff_isGunky`—the Stone dictionary

internalized into the object language) and, more strikingly, is *Henkin-absolute* (`sat_freeUFax_iff_isGunky_henkin`): comprehension forces the would-be refuting witness—the principal up-set of an atom, definable with one parameter—into every general model, so the axiom’s truth value cannot vary across Henkin perspectives. Where the latitude genuinely bites is located exactly: Σ_1^1 truths with non-definable witnesses are *not* family-absolute (`sigma11_not_family_absolute`), while Π_1^1 truths pass downward (`sat_allSO_anti`). And the contrast pole, Dedekind categoricity for full second-order Peano structures, is proved alongside (`dedekind_categoricity`). Appendix A transcribes the statements. These are satisfiability (model-existence) results relative to the ambient metatheory; and reflexively, the ambient type theory plays the role of full semantics for our own formalization—the regress of pinning, visible in the architecture, exactly as predicted.

“Free ultrafilters need the axiom of choice; this betrays the constructive, phenomenological starting point.” A real tension, honestly noted. The `hyperfilter` is classical, and our cited Łoś anchor lives in a classical metatheory. Two things soften the tension without hiding it. First, the *finite, constructive* pole is also formalized: Goertzel’s constructive nonstandard analysis and the topos internal-language view give the coordinate’s constructive self-understanding [6, 12], so the picture has a constructive dual, not only a classical primary. Second, the phenomenology motivates *priority of the continuum*, which is compatible with a constructive treatment of the continuum (smooth infinitesimal analysis, synthetic differential geometry) even where the specific ultrafilter construction is classical. We do not claim to have discharged the tension; we claim to have located it precisely.

“Your own formalism is finitary—ZFC, the empty set, the von Neumann hierarchy, finite formulas, `Finset`—so the program refutes itself.” This is the sharpest objection, and it has a layered reply. *First (use versus mention)*: a finite *description* denotes an infinite *thing*; the finitude of the symbol “ ∞ ”, or of a Lean file, no more makes its referent finite than the brevity of the word “ocean” makes the ocean small. The map is not the territory. *Second (and stronger)*: the *finitary formalism is exactly what the thesis predicts*. Any formal system is a recursive, finitely-presented object—hence, by Principle 1, a principal-ultrafilter shadow, a coordinate from which a perspective points at the One. Our finitary Lean development is therefore not a counterexample to ultrafinitism but an *instance* of it: the instrument a finite perspective uses to gesture at what exceeds it. The thesis is reflexively consistent—it anticipates its own finitary expression. *Third (the honest concession)*: consistency-of-expression is not yet an infinite-native *foundation*, and we do not pretend otherwise. A foundation true to “no finitary elements” would not take the empty set and the finite von Neumann ordinals as primitive atoms. The candidate native homes are real and known: universal-set theories such as Quine’s NF/NFU, in which there is a universal *set*—a totality

that is itself an object, with no forced bottoming-out [25]; non-well-founded and coalgebraic foundations, where Aczel’s anti-foundation axiom permits circular, bottomless structure and identity is fixed by bisimulation (behaviour) rather than by atomic constitution [26]; *atomless* (“gunky”) mereologies, in which every region has proper sub-regions so that there is *no minimal part and no empty individual*—the mereological form of “you can always look deeper,” and the most direct answer to what becomes of the empty set and the foundational atom (they simply do not occur) [31]; and structural foundations (topos theory, homotopy type theory) in which elements are derived from universal properties rather than posited, with the adjoint-modal home of Section 3.1 recovering the discrete from the continuous [12, 27]. Our current ZFC/Lean scaffolding sits at the *principal* pole by necessity of being a tool; the migration of the *foundation* toward these infinite-native settings—and the precise fate of the empty set and the cumulative hierarchy within them—is a genuine open problem, which we flag rather than hide.

“Physics is finite: Planck cutoffs, holographic bounds, finite information.” Each such bound is a *chosen scale*—a coordinate—not a denial of the continuum it cuts. The field formalism keeps the continuum primary and treats the cutoff as a regulator to be sent away or renormalized (Section 8); a finite information bound is a statement about what a situated observer can resolve, i.e. a distinguishability band in the sense of the constructive-nonstandard concept, which is precisely the *principal* pole. The finiteness of physics, where real, is the finiteness of a perspective.

“Granting all this, what does it *buy* you?” Three proved fragments and a method: a transfer semantics for conjecture (Section 5.1), an exact precise-as-principal / imprecise-as-free account of credence (Section 5.2), and a construction-order prescription for stochastic processes that builds the infinite object first (Section 5.3). A reframing that pays in theorems and methods is more than decoration.

10 Honest scope

Remark 18 (what is proved versus what is envisioned). *Proved, sorry-free, and cited above:* Łoś’s theorem and compactness (`models_Uprod`, `compact`); the precision/unanimity and Walley-envelope theorems (`inheritanceCredalWidth_eq_zero_iff`, `lowerInheritanceStrength_isGLB/_isLUB`); the credal scrutability gap and its unit-width indicator; the sampled-vs-full control witness (`flies_mem_winged_scrutability_frontier`); and the de Finetti mixture with its count-only marginals and unique latent measure. The hyperreal identity $\mathbb{R}^* = \text{Germ}(\text{hyperfilter } \mathbb{N}) \mathbb{R}$ is `mathlib`’s. The `UltraTrue/PreciseFamily/OpenFamily` wrapper of Definition 8 is now also proved (principal collapse, invariance, envelopes, dichotomy, free/principal separation, Łoś transfer), as are the two-semantics and Henkin-absoluteness

results of the categoricity reply. *Envisioned, and labelled as such:* the infinite-primary construction of ν as the route to a global carrier; and the replacement of the misnamed `Hyperseed/Ultrainfinitism.lean` perspective shell by the ultraproduct reading. No claim in this essay is dressed beyond what its cited anchor proves.

11 Related work and influences

The mathematical substrate is classical: ultraproducts and Łoś’s theorem from model theory; Robinson’s nonstandard analysis and its hyperreal continuum, with the accessible treatments of Keisler, Goldblatt, and Nelson’s radically-elementary probability [14, 10, 9, 13]. The imprecise side is Walley’s lower/upper previsions and natural extension, meeting exchangeability through de Finetti [16, 3, 4]. The constructional and scrutability heritage is Carnap and Chalmers, with Russell’s maxim of constructions over inferences [1, 2, 15]. The operational ontology and the observer-relative, field-first physics are Goertzel’s Hyperseed and world-crystal lines [6, 7, 8]; the local/global, discrete/continuous duality is the topos internal-language view [12]; and the PLN truth-value machinery our credal theorems sit in is [5]. These are influences and tools, named as such.

12 Conclusion

Ultrainfinitism asks us to reverse a habit of thought. The finite is not the bedrock from which we cautiously approach the infinite; it is the shadow the infinite casts when a perspective picks a coordinate. The thesis is phenomenologically motivated (nothing finite is ever given), mathematically housed (ultraproducts and Łoś), and—crucially—already anchored, in pieces, in proof: Łoś for conjecturing in the limit, the credal-width and envelope theorems for precise-as-principal and imprecise-as-free, and the de Finetti mixture for the infinite-primary construction of stochastic process laws. Conjecture, credence, and concept formation turn out to be one activity seen three ways—construction in the infinite. And the empty-set/atom worry, the sharpest objection to an infinite-native foundation, now has a worked prototype: gunky mereology, where “no atom” is provably “infinite” (Theorem 14), with two concrete atomless algebras and the Stone dictionary tying atoms to principal-ultrafilter points.

The next cycle is core, not application. The formal-core wrapper is now done (`UltraTrue` with principal collapse, invariance, envelopes, the dial dichotomy, free/principal separation, and Łoś transfer—Section 4), so the ladder moves up a rung. First, keep indexing the relative $\text{One}_{(M,U)}$ and isolate what survives change of coordinate, ultrafilter, and family—turning the principal/free dial from an organizing principle toward an adequacy theorem, with Stone duality (Theorem 16) and the dial dichotomy as the first rungs. Second, develop the modal/type-theoretic second pillar—the discrete as an adjoint shadow of a cohesive ground (Section 3.1)—into a genuine alternative home. We deliber-

ately leave the physics lane for after the semantics is harder to wobble. Less manifesto, more core: that is where the One and its finite shadows become a foundation rather than a picture.

A Formalized statements (Lean 4)

The gunky-mereology results of Section 6 are formalized, sorry-free and axiom-clean (only `propext`, `Classical.choice`, `Quot.sound`), in `Mettapedia/Logic/GunkyMereology.lean`. The load-bearing statements verbatim:

```

/-- Gunky = atomless: every non-bottom element has a proper non-
    bottom part. -/
def IsGunky (a : Type u) [PartialOrder a] [OrderBot a] : Prop :=
  forall x : a, x != bot -> exists y, y != bot /\ y < x

/-- Gunk forces the infinite. -/
theorem infinite_of_isGunky [Nontrivial a] (hg : IsGunky a) :
  Infinite a

/-- The continuum is gunky (and hence infinite). -/
theorem isGunky_nnreal : IsGunky NNReal

/-- Boolean witness: clopens of Cantor space are gunky. -/
theorem isGunky_clopens_cantor : IsGunky (Clopens (Nat -> Bool))

/-- Heyting witness: the open regions of the line are gunky. -/
theorem isGunky_opens_real : IsGunky (Opens Real)

/-- Negative example: a two-valued algebra is not gunky (its top is
    an atom). -/
theorem not_isGunky_bool : not (IsGunky Bool)

/-- Stone weld: powerset atoms are singletons = principal-ultrafilter
    points. -/
theorem isAtom_set_singleton (x : a) : IsAtom ({x} : Set a)
theorem not_isGunky_set [Nonempty a] : not (IsGunky (Set a))

```

The full Stone duality (Theorem 16) is formalized in full generality in `Mettapedia/Logic/StoneGunkDuality.lean`, and Sider's argument (Section 6.4) in `Mettapedia/Logic/Metaphysics/SiderNihilism.lean`; both are sorry-free and axiom-clean:

```

/-- The Stone space of a Boolean algebra: the prime spectrum of its
    Boolean ring. -/
abbrev StoneSpace (B) [BooleanAlgebra B] := PrimeSpectrum (AsBoolRing B)

/-- Stone representation: B is (order-iso to) the clopen algebra of
    its Stone space. -/
noncomputable def stoneRepr (B) [BooleanAlgebra B] : B ~o Clopens (
  StoneSpace B)

/-- Stone duality: atomless (gunky) iff perfect Stone space. -/
theorem isGunky_iff_perfect_stoneSpace : IsGunky B <-> PerfectSpace (
  StoneSpace B)

```

```

/-- A "world" is a parthood structure; Sider's nihilism: every object
    is a simple. -/
def SiderNihilism (W : Mereology) : Prop := forall a : W.carrier, a
    != bot -> IsAtom a
def Nec (P : Mereology -> Prop) : Prop := forall W, P W      -- box
def NonContingent (P : Mereology -> Prop) : Prop := Nec P \ / Nec (fun
    W => not (P W))

/-- Gunk excludes nihilism (premise 3). -/
theorem gunky_not_siderNihilism (hg : Gunky W) : not (SiderNihilism W
    )

/-- Sider's argument, valid: granting premise 1, nihilism is
    necessarily false. -/
theorem siders_argument (h1 : NonContingent SiderNihilism) :
    Nec (fun W => not (SiderNihilism W))

/-- But premise 1 fails in the full space of mereologies: nihilism is
    contingent. -/
theorem not_nonContingent_siderNihilism : not (NonContingent
    SiderNihilism)

```

The two-semantics result (the categoricity-rejoinder reply) is formalized in Mettapedia/Logic/Metaphysics/MonadicSecondOrder.lean, .../UltraintinismTwoSemantics.lean, and .../DedekindCategoricity.lean; all sorry-free and axiom-clean:

```

/-- Satisfaction with second-order quantifiers relativized to a
    family S.
    S = univ gives standard (full) semantics; a HenkinFamily gives
    general models. -/
def Sat (S : Set (Set M)) (vF : Fin n -> M) (vS : Fin m -> Set M) :
    MSO n m -> Prop

/-- A Henkin family: an SO-quantifier domain closed under
    comprehension. -/
structure HenkinFamily (M) [BooleanAlgebra M] where
    family : Set (Set M)
    comprehension : ... -- every MSO-definable set (with parameters)
        is in the family

/-- The second-order axiom: every ultrafilter is free (no least
    element). -/
def freeUFax : MSO 0 0 := allSO (imp isUltrafilterFml (not
    hasLeastFml))

/-- Internalized Stone dictionary: over standard semantics, the free-
    ultrafilter
    axiom is equivalent to first-order atomlessness. -/
theorem sat_freeUFax_iff_isGunky :
    SatSentence Set.univ freeUFax <=> IsGunky M

/-- T_UI = {gunkAx, nontrivAx, freeUFax} is satisfiable under
    standard semantics
    (witness: the Cantor clopen algebra) ... -/
theorem T_UI_satisfiable_standard :

```

```

exists W : BAWorld, SatTheory Set.univ T_UI

/-- ... and under Henkin semantics, by the same witness. -/
theorem T_UI_satisfiable_henkin :
  exists W (H : HenkinFamily W.carrier), SatTheory H.family T_UI

/-- Henkin absoluteness: comprehension forces the principal up-set of
    an atom
    (definable with one parameter) into every Henkin family, so the
    axiom's
    truth value agrees with standard semantics in every general model
    . -/
theorem sat_freeUFax_iff_isGunky_henkin (H : HenkinFamily M) :
  SatSentence H.family freeUFax <=> IsGunky M

/-- Hence the theory holds over EVERY Henkin family on the gunky
    witness. -/
theorem T_UI_satisfiable_all_henkin :
  exists W : BAWorld, forall H : HenkinFamily W.carrier, SatTheory H.
    family T_UI

/-- Where latitude bites: Sigma-1-1 truths (non-definable witnesses)
    are NOT
    family-absolute, while Pi-1-1 truths pass downward (
    sat_allSO_anti). -/
theorem sigma11_not_family_absolute :
  SatSentence Set.univ existsUFax /\ exists S, not (SatSentence S
    existsUFax)

/-- The contrast pole: full second-order induction pins the naturals.
    -/
theorem dedekind_categoricity (P Q : PeanoStructure) :
  exists e : P.carrier ~> Q.carrier,
    e P.zero = Q.zero /\ forall x, e (P.succ x) = Q.succ (e x)

```

The formal core of Definition 8 (Mettapedia/Logic/Metaphysics/
 UltrainfinitismCore.lean, sorry-free, axiom-clean), plus the dial-transfer
 pack:

```

/-- Truth from the perspective of an ultrafilter: the verdict of the
    relative One (M, U). Paper name: "ultraintuitively true". -/
def UltraTrue (U : Ultrafilter I) (P : I -> Prop) : Prop :=
  eventually i in U, P i

/-- Principal collapse: the finite is the principal shadow. -/
theorem ultraTrue_pure : UltraTrue (pure i) P <=> P i

/-- Invariance: re-coordinatizing the index re-coordinatizes the One.
    -/
theorem ultraTrue_map : UltraTrue (U.map e) P <=> UltraTrue U (fun i
  => P (e i))

/-- Envelopes: lower/upper over all perspectives = forall/exists
    coordinates. -/
theorem ultraTrue_all_iff : (forall U, UltraTrue U P) <=> forall i
  , P i
theorem ultraTrue_exists_iff : (exists U, UltraTrue U P) <=> exists i

```

```

    , P i

/-- The dial dichotomy: open exactly when not precise, i.e. exactly
when
the coordinates genuinely disagree. -/
theorem openFamily_iff_not_precise : OpenFamily P <-> not (
  PreciseFamily P)

/-- Free vs principal: the free One affirms what a coordinate-shadow
denies. -/
theorem hyperfilter_pure_disagree :
  UltraTrue (hyperfilter Nat) (fun n => n != 0) /\
  not (UltraTrue (pure 0) (fun n => n != 0))

/-- Los transfer, wrapper form; and B is elementarily equivalent to
every ultrapower of itself. -/
theorem ultraTrue_iff_uprod :
  UltraTrue U (fun i => A i |= phi) <-> Uprod A U |= phi
theorem ultrapower_elementary : Uprod (fun _ => B) U |= phi <-> B |=
  phi

/-- Transfer pack (StoneGunkDuality.lean): move the atomic/gunky dial
to
the right carrier (isomorphs, parts). -/
theorem isGunky_congr (f : A ~o B) : IsGunky A <-> IsGunky B
theorem isGunky_Iic (hg : IsGunky A) (a : A) : IsGunky (Set.Iic a)
theorem perfectSpace_stoneSpace_congr (f : B ~o C) :
  PerfectSpace (StoneSpace B) <-> PerfectSpace (StoneSpace C)

```

The native ultrapower/Łoś anchor (`models_Uprod`) lives in `Mettapedia/Logic/Foundation/Foundation/FirstOrder/Ultraproduct.lean`; the credal-width, Walley-envelope, scrutability-gap, and de Finetti statements are cited at their source lines in the body.

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